

**CROSS-HEDGING CASTOR MEAL PRICE RISK USING
NCDEX CASTOR SEED FUTURES**

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1. Objective

Castor meal has no futures contract of its own on NCDEX, so a holder of castor meal cannot hedge it directly. This project tests whether NCDEX castor seed futures, a related commodity, can be used instead. This is a cross-hedge. The project estimates the hedge ratio that minimises risk, measures how much risk it removes over the estimation period (in-sample) and then checks whether the same hedge still works on a later period it has not seen (out-of-sample).

2. Assets

Asset A: castor meal. Castor meal is what remains after oil is extracted from castor seed. The price is the daily quote published by The Solvent Extractors' Association of India (SEA) for "Castorm meal Ext (FOR) (Bulk) (Ex-Kandla)", in rupees per tonne.

Asset B: castor seed futures. NCDEX contract CASTOR, delivery at Deesa, one contract expiring each month around the 20th, priced in rupees per quintal.

The two prices are linked because castor seed is crushed to produce castor oil and castor meal, but the link is loose: meal has its own demand, and much of the seed's value goes into the oil. NCDEX lists futures on castor seed and refined castor oil but has no castor meal contract, so a direct hedge is not available.

3. Data and sample creation

	Castor meal (spot)	Castor seed (futures)
Source	SEA daily quotes	NCDEX daily bhavcopy files
Price used	Published quote	Closing price of each contract
Unit	Rs per tonne	Rs per quintal
Period	25 Sep 2023 to 28 Sep 2026	25 Sep 2023 to 28 Sep 2026
Observations	546 usable quotes (of 647 posts)	744 trading days

Table 1. Data. SEA: seaofindia.com/seaquotes. NCDEX: ncdex.com/markets/bhavcopy. Accessed 23 to 29 September 2026.

Every NCDEX file was checked to confirm it belongs to the date it was downloaded for, and every SEA quote was compared again with the live SEA web page; all values matched. The castor meal prices are SEA figures, not NCDEX prices.

How the dates are aligned. A daily return is used only when two consecutive NCDEX trading days both have an SEA quote. Weekends and holidays are skipped. No prices are estimated, filled in or copied forward, because that would create artificial price movements. This gives **496 daily returns** from 26 Sep 2023 to 28 Sep 2026.

Data cleaning. Of the 647 daily SEA posts in the period, 546 give a usable castor meal price and are used

. The remaining posts were set aside under fixed rules (for example, a date that does not match the publication date, a duplicate of an earlier post or an obvious typing error). The SEA price is a reported market rate and often stays the same from one day to the next (about one day in three).

4. Building a continuous futures price

Each futures contract expires, so three years of data must be pieced together from 36 monthly contracts. The course reading *Continuous Contracts Explained* identifies two problems. First, if a series simply switches contracts at expiry (the "spot-month" method), it holds contracts in their final weeks when trading is thin and delivery can be forced (p.1). Second, joining contracts creates artificial jumps, because each month trades at its own price; these jumps are not real market moves (p.6 to 7).

The project handles these as follows:

- **When to switch (from the paper, p.2).** Move to the next contract once its open interest becomes larger than the current contract's, that is, when the market has moved there.
- **No use of future information (my adjustment).** Open interest is only published after the close, so the switch happens at the next day's close.
- **Safety limit (my adjustment).** If the switch has not happened with 7 trading days left, it is made anyway. NCDEX castor seed is settled by compulsory delivery and its last 3 trading days are a delivery (tender) period. This limit was needed for 4 of the 36 rolls.
- **No artificial jumps (paper, p.7 to 9).** Each daily futures return is calculated on the same contract on both days. The price gap on a switch day is never counted as a return. A back-adjusted series, in which each gap is removed from earlier prices as the paper describes, is kept in the workbook for reference.

Table 2 shows three of the 36 switches as examples. On the switch day the old and new contracts have different prices; this difference (the gap) is not a market movement and is never counted as a return.

Switch date	Reason	Old to new contract	Old / new price	Gap
4 Oct 2023	Open interest moved	Oct 2023 to Nov 2023	6,135 / 6,193	+58
9 Nov 2023	7 days left (limit)	Nov 2023 to Dec 2023	5,953 / 5,873	-80
3 Sep 2026	Open interest moved	Sep 2026 to Oct 2026	7,542 / 7,659	+117

Table 2. Example contract switches (prices in Rs per quintal). All 36 switches are listed in the Roll Schedule sheet of the workbook.

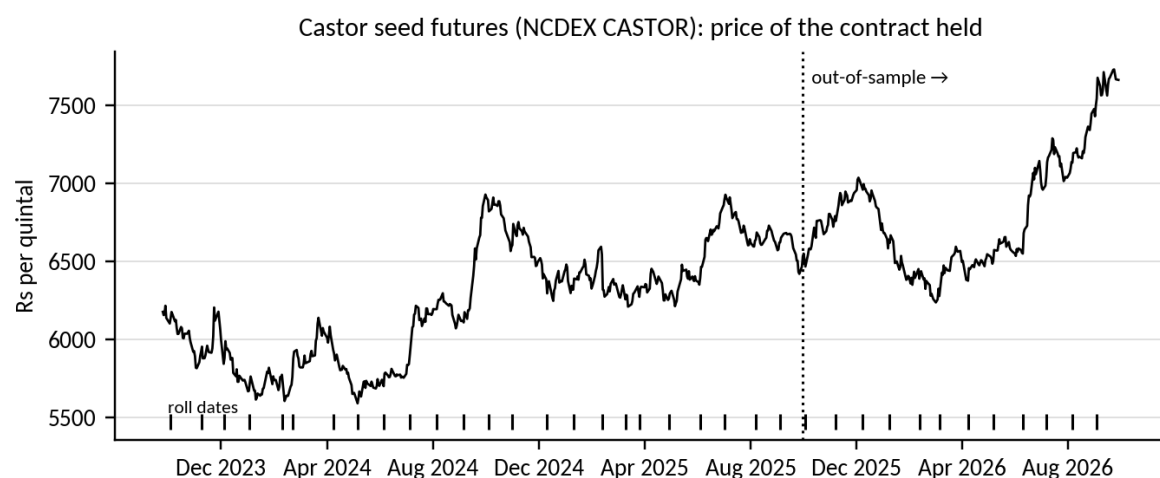


Figure 1. Closing price of the castor seed contract held each day. Tick marks show the 36 roll dates. Small steps at the ticks are contract changes and are not used as returns. The dotted line marks the start of the out-of-sample period.

As a check, other switching rules were tested on the same data. Switching 10 days before expiry, or on volume instead of open interest, gave almost the same results. Holding each contract until expiry gave much weaker results (out-of-sample effectiveness 3.0%), which supports the paper's warning about the final weeks of a contract.

5. Method

All formulas follow Buyukkara, Kucukozmen and Uysal (2022). The symbols are: S_t = castor meal price on day t (SEA), F_t = price of the futures contract held on day t (NCDEX), and $t-1$ = the previous trading day. The futures price is always taken from the same contract on both days.

Daily returns (percentage change, measured in logs):

$$r_{S,t} = \ln(S_t / S_{t-1}) \quad (1) \quad r_{F,t} = \ln(F_t / F_{t-1}) \quad (2)$$

Hedged return. A holder of castor meal sells futures worth h times the value of the meal. The futures loss or gain offsets part of the meal price change:

$$r_{H,t} = r_{S,t} - h \times r_{F,t} \quad (3)$$

The hedge ratio h is set in three ways:

- **OLS (constant ratio).** Regress castor meal returns on futures returns: $r_{S,t} = a + h \times r_{F,t} + e_t$ (4). The slope h is the ratio that gives the lowest variance of the hedged return, equal to $\text{Cov}(r_S, r_F) / \text{Var}(r_F)$. It is estimated once and then kept fixed.
- **Naïve.** $h = 1$: sell futures of the same value as the meal held. Used as a simple benchmark.
- **GARCH (daily ratio).** A diagonal VEC GARCH(1,1) model updates each day the variance of futures returns ($H_{FF,t}$) and the covariance between meal and futures returns ($H_{SF,t}$), using the previous day's price shocks and the previous day's values. The hedge ratio for day t is $h_t = H_{SF,t} / H_{FF,t}$ (5), so it changes every day.

Hedge effectiveness is the share of the castor meal price risk (variance) removed by the hedge:

$$HE = 1 - \text{Var}(r_H) / \text{Var}(r_S) \quad (6)$$

For example, $HE = 0.07$ means the hedged position has 7% less variance than holding castor meal unhedged. Variances are sample variances of the daily returns.

In-sample and out-of-sample. The data are split by date. The hedge ratios are estimated only on 26 Sep 2023 to 26 Sep 2025 (275 returns). They are then applied, unchanged, to 1 Oct 2025 to 28 Sep 2026 (221 returns). The GARCH model is also estimated only on the first period and uses only past data each day.

6. Results

The returns. Castor meal returns are more than twice as volatile as the futures returns (daily standard deviation 1.60% against 0.67%) and have many unchanged days followed by occasional large jumps. Both return series are stationary. The two price levels do not move together in the long run (no cointegration), so the GARCH model uses no long-run correction term.

Return diagnostics

Statistic	Castor meal spot	Castor seed futures
Matched return observations	496	496
Standard deviation	1.597%	0.668%
Skewness	0.318	0.015
Excess kurtosis	4.882	1.237
Jarque–Bera p-value	<0.001	<0.001
Ljung–Box Q,p-value	0.045	0.248
ADF p-value, returns	<0.001	<0.001

Model	Hedge ratio	HE in-sample	HE out-of-sample
OLS	0.614	6.79%	6.59%
Naive	1.000	4.10%	4.26%
GARCH	new each day: 0.07 to 1.24 (average 0.51)	6.99%	7.02%

Table 3. Hedge ratios and hedge effectiveness. In-sample n = 275, out-of-sample n = 221.

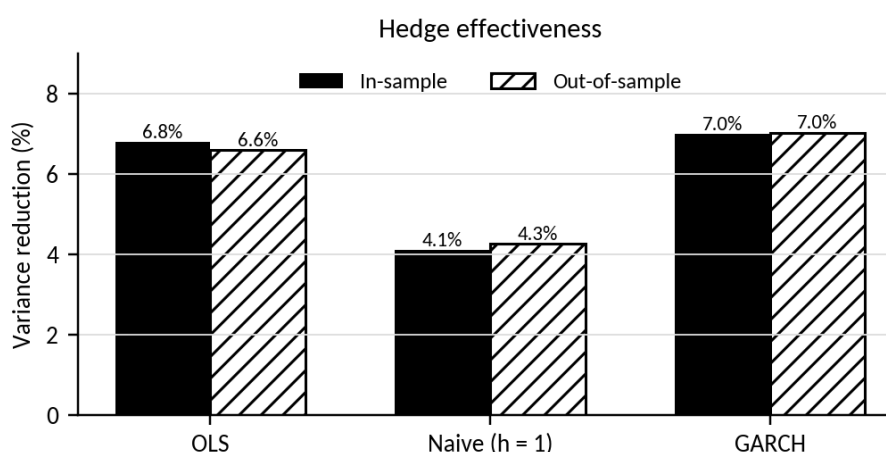


Figure 2. Share of castor meal price risk removed by each hedge.

What the numbers mean. The OLS hedge ratio of 0.614 means selling futures worth about 61% of the value of the castor meal held. The ratio is statistically significant ($t = 4.46$), so the link between the two prices is real. It is also weak: the correlation between daily returns is only 0.26, and the hedge removes under 7% of the risk, leaving about 93% in place.

The in-sample and out-of-sample results are very close for every model, so the hedge did not fall apart on new data. The naive hedge does worse because selling one-for-one is too much for such a loose relationship. GARCH is slightly better than OLS (by 0.2 to 0.4 percentage points), but the difference is too small, and the model's estimates too uncertain, to prefer it over the simpler OLS hedge.

7. Limitations

- The waiting day and the 7-day safety limit in the roll rule are my own choices, added to the method in the paper.
- The GARCH model is estimated once on the first period and is not re-estimated during the test period.
- Trading costs, margins and whole-contract sizes (5 tonnes per contract) are ignored, so a real hedge would do slightly worse.

8. Conclusion

Castor seed futures provide only a small hedge for castor meal. The best simple hedge (OLS, ratio 0.614) removes about 6.8% of daily price risk in-sample and 6.6% out-of-sample, and this result is stable across the two periods. A more complex GARCH hedge adds very little. The main lesson is that a cross-hedge can only remove the risk the two commodities share, and for castor meal and castor seed that shared part is small. Careful construction of the futures series mattered: switching contracts before the delivery period and never counting switch-day gaps as returns gave clearly better and more reliable results than simply holding each contract to expiry.

References

- Buyukkara, G., Kucukozmen, C. C. and Uysal, E. T. (2022) 'Optimal hedge ratios and hedging effectiveness: An analysis of the Turkish futures market', *Borsa İstanbul Review*, 22(1), pp. 92–102.
- Continuous Contracts Explained* (n.d.) Course reading (author and date not stated).
- NCDEX (2026) *Bhav Copy and Contract Specifications: Castor Futures Contract*. Available at: <https://www.ncdex.com> (Accessed: 29 September 2026).
- SEA, The Solvent Extractors' Association of India (2023–2026) *SEA Quotes*. Available at: <https://seaofindia.com/seaquotes/> (Accessed: 29 September 2026).
- All calculations are in the accompanying workbook, *Castor_Meal_Cross_Hedge_FINAL.xlsx*.